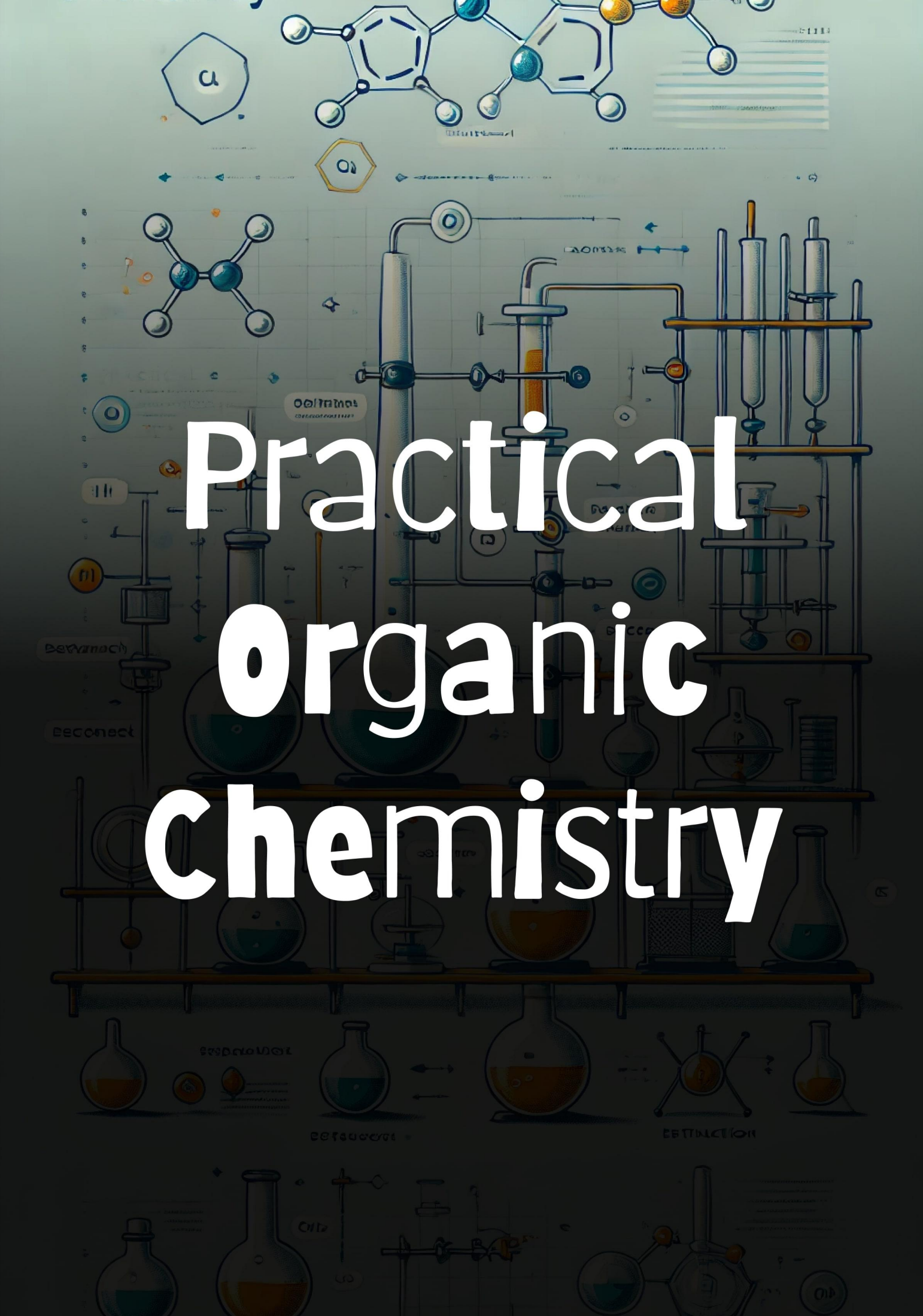

Practical
Organic
Chemistry

In this section, we study about organic compounds having different elements like C, S, halogen, H, P etc.

## Test for Hydrogen and Carbon

Organic substance is passed through CuO at high temperature. $CO_2$, $H_2O$ will produce if organic substance contains carbon & hydrogen respectively. If $CO_2$ is passed through lime water ($Ca(OH)_2$), it becomes milky due to formation of $CaCO_3$.

If $H_2O$ is present, it is passed through anhydrous $CuSO_4$, & it gives blue to formation of blue vitrial" ($CuSO_4 \cdot 5H_2O$)

($CuSO_4 \cdot 5H_2O$)

$$C + 2CuO \longrightarrow CO_2 + 2Cu$$

$$2H + CuO \longrightarrow H_2O + Cu$$

$$CO_2 + Ca(OH)_2 \longrightarrow CaCO_3 \downarrow + H_2O$$

$$5H_2O + CuSO_4 \longrightarrow CuSO_4 \cdot 5H_2O$$

(white) (blue vitrial)

## Test for Nitrogen, Sulphur and Halogen

The test is called "Lassaigne's Test"

In this process, organic substance is heated in an ignition tube in presence of sodium metal and we will get "sodium extract" (ionic in nature).

$$Na + C + N \longrightarrow NaCN$$

$$Na + X \longrightarrow NaX$$

$$2Na + S \longrightarrow Na_2S$$

## i Test for Nitrogen

- Sodium extract is passed through ferrous sulphate solution in presence of concentrated sulphuric acid. "Prussion Blue" colour will obtain due to presence of ferri ferro cyanide.

$$NaCN + FeSO_4 \longrightarrow Na_4[Fe(CN)_6]$$

$$[Fe(CN)_6]^{-4} + Fe^{+3} \longrightarrow Fe_4[Fe(CN)_6]_3$$

($Fe^{+3}$: same Fe is oxidised; $Fe_4[Fe(CN)_6]_3$: (Prussion blue))

## ii TEST FOR SULPHUR

a. Sodium extract is passed through lead acetate in presence of acetic acid. Black precipiate will be obtained due to formation of PbS.

$$Na_2S \xrightarrow[Pb(CH_3COO)_2]{CH_3COOH} PbS \downarrow \text{Black}$$

b. Sodium extract is passed through sodium nitro prussid, violet colour will be obtained.

$$S^{2-} + N_2[Fe(CN)_5NO] \longrightarrow Na_2[Fe(CN)_5NO_5]$$

($Na_2[Fe(CN)_5NO_5]$: (Violet))

## iii TEST FOR HALOGEN

- Sodium extract is passed through silver nitrate ($AgNO_3$) in presence of nitric acid ($HNO_3$). If precipitate is obtained there will be a halogen.

$$AgNO_3 + SE \xrightarrow{HNO_3} AgX \downarrow$$

($AgX \downarrow$: precipitate)

If white precipitate which is soluble in aqueous ammonia solution it confirms that "halogen is chlorine".

$$AgCl \xrightarrow{NH_4OH} \text{soluble}$$

(AgCl: white)

If yellow precipitate which is partially soluble in aqueous ammonia solution it confirms that halogen is bromines.

$AgBr \xrightarrow{NH_4OH}$ Partially soluble

(AgBr: yellow)

If yellow precipitate which is insoluble in aqueous ammonia sol, then iodine is confirmed.

$AgI \xrightarrow{NH_4OH}$ insoluble

## iv TEST FOR PHOSPHORUS

- organic compound is passed through sodium proxide, all phosphorus will be converted into phosphate which is passed through $HNO_3$ to give phosphoric acid which on passing through ammonium molybdate. It will give yellow colour of ammonium phosphate molybdate.

organic compound $\xrightarrow{Na_2O_2}$ $Na_3PO_4$ $\xrightarrow{HNO_3}$ $H_3PO_4$ $\xrightarrow{(NH_4)_2MoO_4}$ $(NH_4)_3PO_4.12\,MoO_3$ (yellow)

# QUANTATIVE ANALYSIS

In this section we will calculate percent composition of elements present in organic compounds.

## Leibig's Method / combustion Method

- W is mass of organic substance.
- $W_1$ is mass of $Co_2$ obtained after combustion.

→ $W_2$ is mass of $H_2O$ obtained after combustion.

$$\underset{w}{C_xH_y} + \left(x - \frac{y}{4}\right)O_2 \longrightarrow \underset{W_1}{xCO_2} + \frac{y}{2}\cdot \underset{W_2}{H_2O}$$

$$\%C = \frac{W_1}{44} \times \frac{12}{W} \times 100\%$$

$$\%H = \frac{W_2}{18} \times \frac{2}{W} \times 100\%$$

Aqueous KOH absorbs $CO_2$ Anhydrous $CaCl_2$, absorbs $H_2O$,

Mass of $CO_2$ is obtained due to weight increase in aqueous KOH tube.
Mass of $H_2O$ is obtained from weight increase in anhydrous $CaCl_2$ tube.

POC

▷ Analysis of Nitrogen

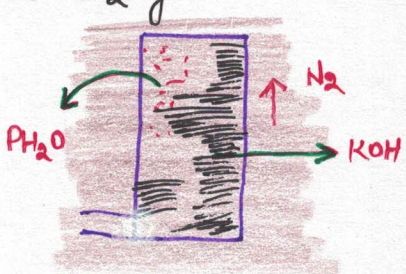
There are two methods for determining % nitrogen in organic compounds

1. DUMA's method
2. KJELDAHL's method

## 1 DUMA'S METHOD (DM)

In this method, organic substance is burnt in presence of CuO, it'll give $CO_2$, $N_2$, $H_2O$ and copper metal. These products are collected in a glass tube containing aqueous KOH solution. $N_2$ gas will be collected over the solution. so, at that temperature there will be some vapour pressure. so pressure due to $N_2$ gas –

$$P_{N_2} = P_{moist} - \text{aqueous tension}$$




where aq. tension is vapour pressure of water at that temperature.

$$C_2H_yN_2 + (2x + y/2)\,CuO \longrightarrow xCO_2 + \frac{y}{2}H_2O + \frac{z}{2}N_2 + (2x + y/2)\,Cu$$

For volume of $N_2$, there will be marking on glass tube. Now with the help of $PV=nRT$ we can calculate moles of $N_2$ gas.

Hence $\% N = \left(\frac{n_{N_2} \times 28 \times 100}{w}\right)\%$

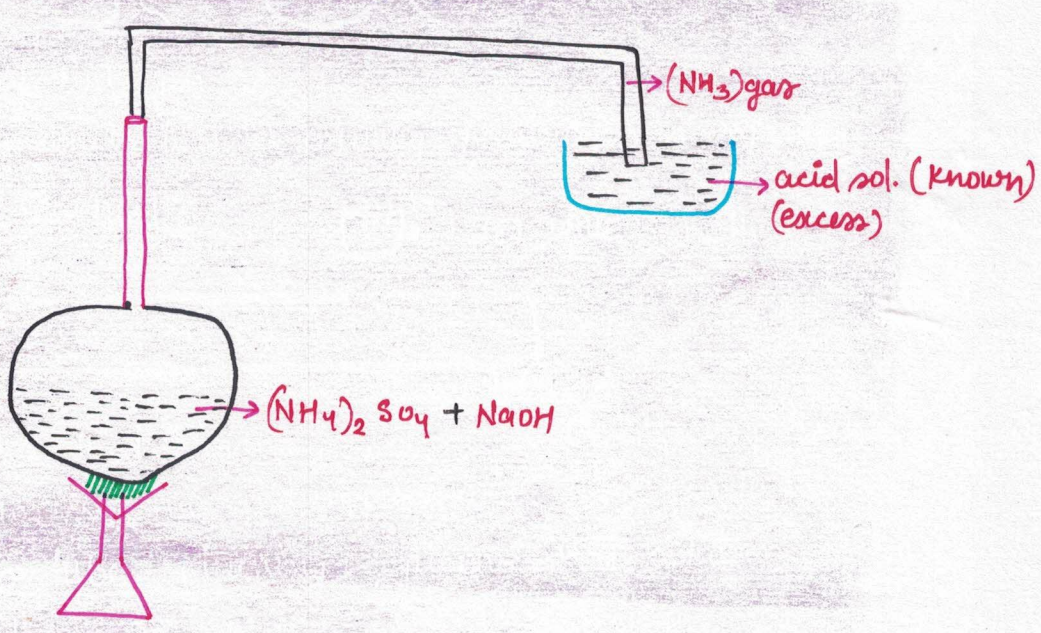
$w$ = Total weight of organic substance

$n$ = moles of $N_2$ gas

## 2. KJELDAHL's METHOD

An organic compound is digested in a closed flask in presence of sulphuric acid & copper sulphate ($H_2SO_4 + CuSO_4$)
Then this flask is added to a distillation column before addition of column NaOH is added. In product we will get ammonia gas.

$$\text{Organic compound} \xrightarrow[CuSO_4]{H_2SO_4} (NH_4)_2SO_4 \xrightarrow{2NaOH} 2NH_3 + Na_2SO_4 + H_2O$$




### IMPORTANT

Ammonia gas is collected (absorbed) in excess acid solution (known). Excess amt. of acid is neutralised with known basic solution.
This is called "Black Titration".

So, equivalent of ammonia = Moles of ammonia because valence factor is 1.

If moles of ammonia = $n$

$$\%N = \left(\frac{n_{NH_3} \times 14 \times 100}{10}\right)\%$$

$W$ = mass of organic compound.

## ANALYSIS OF SULPHUR AND HALOGEN

### FOR RALOGENS

**(i) Carius Method** – In this method, organic substance is passed through fuming $HNO_3$ in presence of $AgNO_3$ and $AgX$ precipitate will be formed.

Organic compound $\xrightarrow[AgNO_3]{\text{fuming } HNO_3}$ $AgX$

$W$ $\qquad$ $W_1$

For $AgCl$ :- $\left(\frac{W_1}{143.5} \times \frac{35.5}{W} \times 100\right)\% \longrightarrow Cl\%$

$AgBr$ :- $\left(\frac{W_1}{188} \times \frac{80}{W} \times 100\right)\% \longrightarrow Br\%$

$AgI$ :- $\left(\frac{W_1}{235} \times \frac{127}{W} \times 100\right)\% \longrightarrow I\%$

### FOR SULPHUR

**(i) Carius Method** – organic substance is passed through fuming $HNO_3$ to get $H_2SO_4$. $H_2SO_4$ is then passed through $BaCl_2$ to form $BaSO_4$ which precipitates.

Organic compound $\xrightarrow{\text{fuming } HNO_3}$ $H_2SO_4$ $\xrightarrow{BaCl_2}$ $BaSO_4\downarrow$

$W$ $\qquad$ $W_1$

$$\%S = \left(\frac{W_1}{233} \times \frac{32}{W} \times 100\right)\%$$

## V TEST FOR OXYGEN

organic substance is reated which gives $O_2$. But $O_2$ is absorbed by carbon to form carbon monoxide. CO can be passed through $I_2O_5$ to form $CO_2$.

(organic compound) $\xrightarrow{\Delta}$ $O_2$ + other compound

$[O_2 + 2C \longrightarrow 2CO] \times 5$

$[5CO + I_2O_5 \longrightarrow 5CO_2 + I_2] \times 2$

$W_1$

It can be seen that 5 mole $O_2$ is giving 10 mole $CO_2$.

So, 1 mole $O_2$ $\longrightarrow$ 2 mole $CO_2$

$$\% O = \left(\frac{W_1}{44} \times \frac{16}{W} \times 100\right) \%$$

"16" is written because 1 mole $O_2$ $\longrightarrow$ 2 mole $CO_2$

## MOLECULAR WT. DETERMINATION FOR ORGANIC COMPOUNDS

### (i) Cryoscopic Method

– when non volatile solute is added to volatile solvent, then vapour pressure of solution will decrease, boiling point will increase and freezing point will decrease.

$\Delta P \downarrow \propto$ B.P $\uparrow$

$\Delta T_B$ = boiling point
$\Delta T_f$ = freezing point

$\Delta T_f \propto$ molality

$\Delta T_f = K_f \times$ molality

W gram solvent & w gram solute.

M → molecular wt. of organic compound (solute).

$\Delta T_f$ → depression in freezing point

$$M = \frac{W \times 1000}{\text{Molality} \times W} = \frac{10 \times 1000}{\frac{\Delta T_f}{K_f} \times W}$$

$$M = \frac{K_f \times W \times 1000}{\Delta T_f \times W}$$

## (ii) Ebullioscopic Method

$$M = \frac{K_b \times 10 \times 1000}{\Delta T_B \times W}$$

same method
($T_B \propto$ molality)

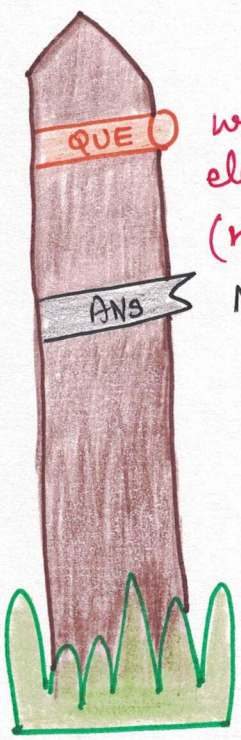
$K_B$ = ebullioscopic constant or metal elavation constant

$W$ = weight of solute

$W$ = weight of solvent

$\Delta T_B$ = elevation in B.P

$M$ = Molecular wt. of organic compound




when one gram organic substance is added to 10 gram water, elevation in boiling point is 2°C.
(molal elevation constant for water is 0.52)

Molality $\frac{1}{M}$ $\frac{100}{10}$ $\frac{100}{M}$

2 $\frac{100}{M}$ 0.52

$M$ $\frac{100 \quad 0.52}{2}$

26 gram Ans.....

## (iii) Silver Salt Method

– It is used for determining molecular weight of organic acids. (R–COOH)

$$R COOH \longrightarrow R COOMg_{W_1}$$

$$R COOAg_{W} \longrightarrow Ag_{W_1}$$

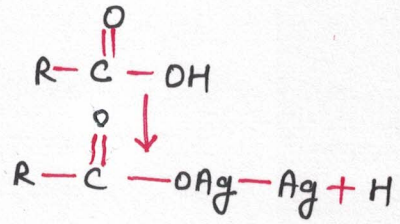
Equivalents will be equal.

$$\frac{W}{E} = \frac{W_1}{108}$$

E will be known.

$E - 180 + 1$ = Equivalent wt. of acid

$$E - 107 = \frac{M}{V \cdot E}$$

As 

$E - 108 + 1$

> Molar mass of acid = $(E - 107) \times V \cdot F$

## (vi) Platinum Salt method ($RNH_2$) (base)

It is used for determining molecular wt. of organic base.

B is any base.

$$2B + 2HCl \longrightarrow B_2H_2Cl \longrightarrow B_2H_2{}^{++}Cl \longrightarrow \underset{W_1}{Pt.}$$

$195 = E_3$

Eq. will be same.

$$\frac{W}{2B \quad 410} = \frac{W_1}{195}$$

B is eq. wt. of base

> Molar mass $B \times V \cdot F$

> No. of equivalents = $\frac{\text{weight}}{\text{equivalent wt.}}$

# Victor Meyer's Test

This is for volatile organic substances.

$$PV = nRT$$

$$PV = \frac{WRT}{M}$$

> $M = \frac{WRT}{PV}$

when volatile substance is heated, it becomes "vapour" and follows ideal gas equation.

Organic substance ignite at high temperature & convert to gas. so, we can use ideal gas equation.

# DEGREE OF UNSATURATIONS

$C_x H_y N_z O_p X_q$

$$D.O.U = (x+1) - \left(\frac{y+q-z}{2}\right)$$

Q. $C_2H_2$

$D.O.U = (2+1) - \frac{(2)}{2} = 3-1 \Rightarrow 2$

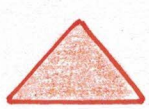
Q. $C_2H_4$

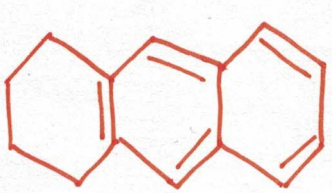
$D.O.U = (2+1) - \left(\frac{4}{2}\right) = 3-2 \Rightarrow 1$

DEGREE OF UNSATURATION – No. of $\pi$ bonds present.

if D.U=1, either 1 double bond or cyclic compound is present.

E.g. $C_3H_6 \Rightarrow (3+1) - \frac{6}{2} = 1$

$(CH_3 - CH = CH_2)$ or 

Q. 

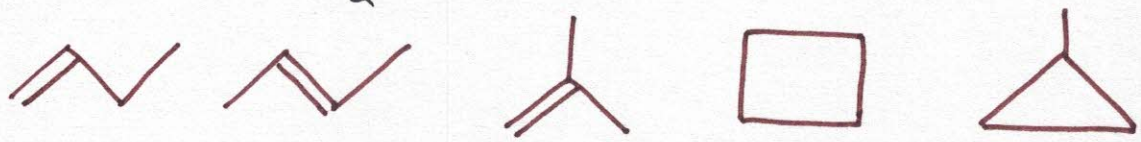
D.U ⇒ No. of $\pi$ bond + No. of cycles

⇒ 5 + 3

⇒ 8

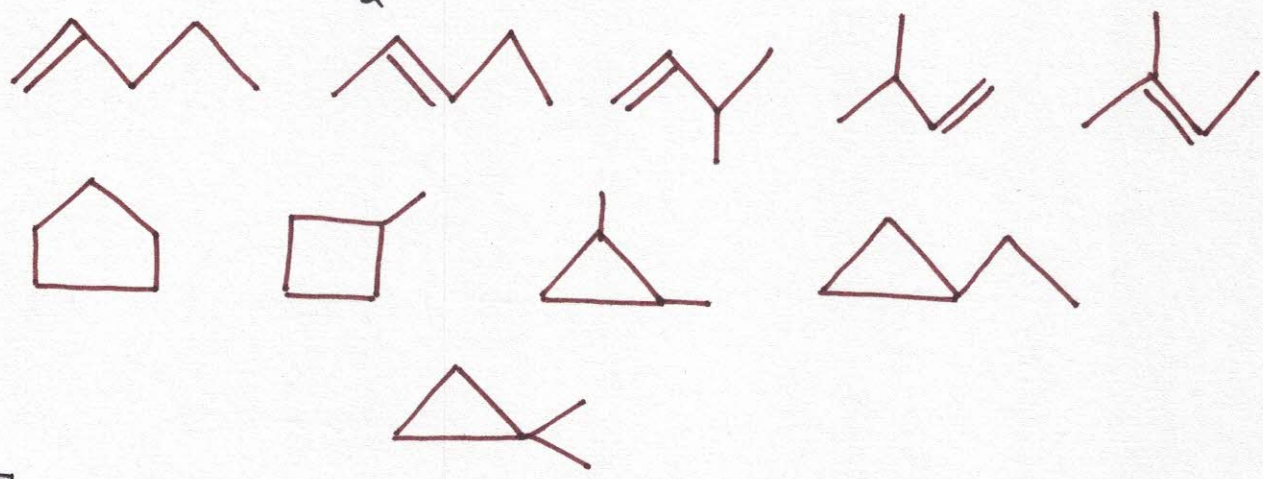
Q. No. of isomers of $C_4H_8$

ANS → D.U $= (4+1) - \frac{8}{2} = 1$



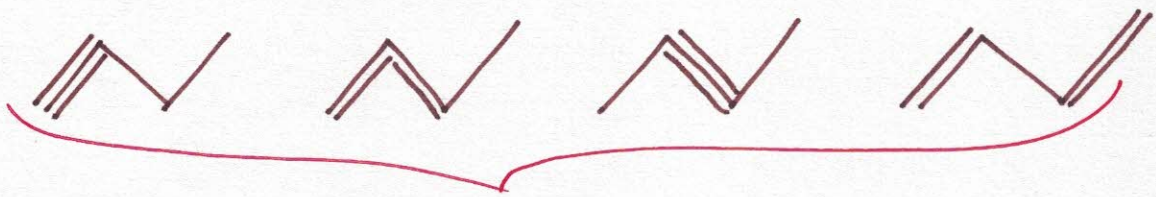
Q. $C_5H_{10}$

ANS → D.U $= (5+1) - \frac{10}{2} = 1$

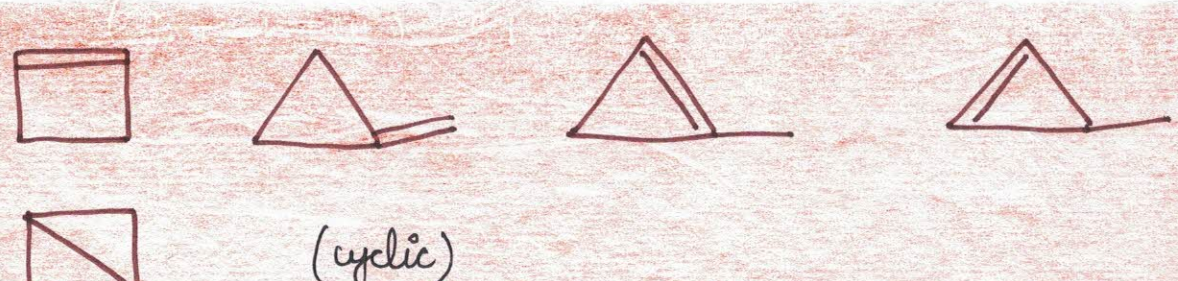


Q. $C_4H_6$

ANS → $(4+1) - \frac{6}{2} \Rightarrow 5 - 3 \Rightarrow 2$

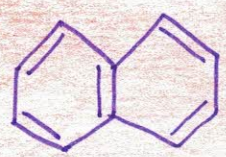


(acyclic)



(cyclic)

Q.



ANS → D.U = 5 double bonds + 2 cyclic ⇒ 7